SESSION 02

# Bank Branch Security Lab

Phase 2 · Identity: Active Directory · October 8, 2026

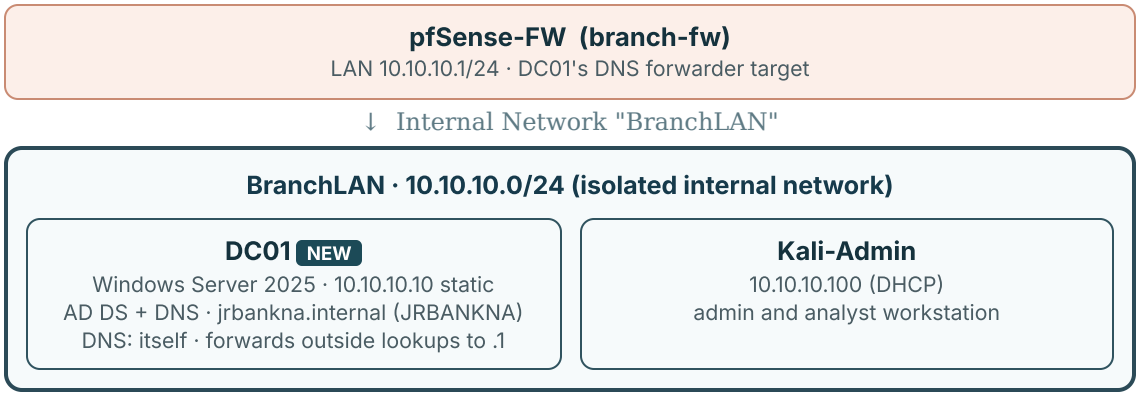
| | |
|---|---|
| **Goal** | Give the branch an identity system: a Windows Server domain controller running Active Directory and DNS, organized the way a real branch is staffed. |
| **Outcome** | Domain controller deployed, a new forest created, DNS forwarding through the firewall, and branch OUs, users and role-based security groups in place. |
| **Systems built** | DC01 (Windows Server 2025 Standard Evaluation) · AD forest jrbankna.internal |
| **Branch network** | Unchanged: 10.10.10.0/24 · DC01 static at 10.10.10.10 |
| **Skills** | Windows Server deployment, Active Directory Domain Services, DNS, OU design, identity and access management, least privilege, PowerShell |

## Overview

This phase added identity to the branch. I built a Windows Server 2025 domain controller, DC01, on BranchLAN with a static address of 10.10.10.10 in the reserved server range. I installed Active Directory Domain Services and promoted DC01 to the first domain controller of a new forest, **jrbankna.internal** (NetBIOS JRBANKNA). DC01 answers DNS for the domain and forwards every other lookup to the pfSense resolver. I then organized the domain like a real branch: a Branch01 OU tree, staff accounts including a separate privileged admin account, and three role-based security groups.

# Architecture

The perimeter is unchanged. DC01, the first server in the reserved range, now provides identity and name resolution for the branch.




## Addressing and DNS

| Item | Value | Notes |
|---|---|---|
| DC01 | 10.10.10.10 / 24 (static) | First server in the reserved range; domain controller and DNS |
| DC01 DNS server | 127.0.0.1 and ::1 | Itself, set automatically when the server was promoted |
| DNS forwarder | 10.10.10.1 (branch-fw) | Lookups outside jrbankna.internal go to the pfSense resolver |
| AD domain | jrbankna.internal · JRBANKNA | .internal is reserved for private networks |

## Domain configuration

| | |
|---|---|
| **Deployment** | Add a new forest (first domain controller) |
| **Root domain** | jrbankna.internal · NetBIOS JRBANKNA |
| **Functional level** | Windows Server 2025 for both forest and domain |
| **DC capabilities** | DNS server and Global Catalog; not read-only |
| **DNS forwarder** | 10.10.10.1 · validated |

# Directory design

```
jrbankna.internal
 ├─ Domain Controllers   (built in · DC01 stays here)
 └─ Branch01
    ├─ Staff
    │  ├─ Tellers
    │  ├─ Managers
    │  └─ IT
    ├─ Admins
    ├─ Workstations
    └─ Groups
```

Every OU was created with "Protect container from accidental deletion" enabled.

## User accounts

| OU | Display name | Logon name | Member of |
|---|---|---|---|
| Staff › Tellers | Wade Wilson | `wade.wilson` | GG-Tellers |
| Staff › Tellers | James Howlett | `james.howlett` | GG-Tellers |
| Staff › Managers | Jean Grey | `jean.grey` | GG-Managers |
| Staff › IT | JR Younger | `jr.younger` | GG-IT |
| Admins | JR Younger (Admin) | `adm.jryounger` | Domain Admins |

Staff display names are public-safe placeholders. Daily work happens on jr.younger; domain changes only on adm.jryounger.

## Security groups

| Group | Scope · type | Members |
|---|---|---|
| GG-Tellers | Global · Security | Wade Wilson, James Howlett |
| GG-Managers | Global · Security | Jean Grey |
| GG-IT | Global · Security | JR Younger |

## What I built

1. Provisioned DC01 (Windows Server 2025 Standard, Desktop Experience) on BranchLAN only, so it reaches the internet exclusively through the firewall.
2. Assigned the static address 10.10.10.10/24, gateway 10.10.10.1, and set the time zone.
3. Installed the Active Directory Domain Services role with Group Policy Management and the AD admin tools.
4. Promoted DC01 to a domain controller for a new forest, jrbankna.internal, with DNS and Global Catalog.
5. Configured a DNS forwarder to 10.10.10.1 and validated it.
6. Built the Branch01 OU tree, created five user accounts, and added the separate admin account to Domain Admins.
7. Created three Global security groups and assigned members by role.
8. Verified accounts and memberships with `Get-ADUser` and `Get-ADGroupMember`.

### Design decisions

| Decision | Rationale |
|---|---|
| Domain controller before workstations | A computer can't join a domain that doesn't exist. Identity is infrastructure others depend on. |
| Static 10.10.10.10 in the reserved range | Every domain member finds the DC through DNS and talks to it for logins. Its address can never change or collide with a DHCP lease. |
| Domain name jrbankna.internal | .internal is reserved for private use and will never exist publicly. Renaming a domain later is painful, so it was settled up front. |
| DNS installed with AD | Active Directory depends on DNS: clients locate domain controllers by records the DC registers in its own zone. |
| Forwarder to pfSense, not root hints | Keeps one controlled exit for DNS. Outside lookups pass through the firewall's resolver, so they can be filtered and logged in one place. |
| Custom OUs, not built-in containers | The built-in Users and Computers containers can't have Group Policy linked to them. OUs can. |
| Separate admin account | Daily work happens on jr.younger; domain changes only on adm.jryounger. A phished everyday account doesn't hand over Domain Admin. |
| Access through groups, GG- prefix | Permissions go to groups, never to individuals. Onboarding and offboarding become group changes. |

## Verification

| Check | How | Result |
|---|---|---|
| Static address and server name | Server Manager › Local Server | Pass · DC01 · 10.10.10.10 |
| Names resolve through the firewall | `nslookup microsoft.com` | Pass · answered by branch-fw |
| AD DS role installed | Add Roles and Features result | Pass · promoted on DC01 |
| Forwarder validates | DNS Manager › Forwarders | Pass · 10.10.10.1 |
| Domain exists | `Get-ADDomain` | Pass · jrbankna.internal · JRBANKNA |
| DC answers for its domain | `nslookup jrbankna.internal` | Pass · 10.10.10.10 via ::1 |
| Five users in Branch01 | `Get-ADUser -SearchBase …` | Pass · all five listed |
| Teller group membership | `Get-ADGroupMember GG-Tellers` | Pass · Wilson, Howlett |
| Domain Admins kept small | `Get-ADGroupMember "Domain Admins"` | Pass · Administrator, adm.jryounger |

## Troubleshooting

### No Windows Server installer in the ISO folder

| | |
|---|---|
| **Cause** | The Evaluation Center lists a "Languages and Optional Features" add-on (SERVER_LOF_PACKAGES) beside the installer. The add-on had been downloaded instead of the installer. |
| **Fix** | Downloaded the correct evaluation ISO and built the VM in parallel. Read the file name before building; edition strings tell you what the file actually is. |

### Kali could not ping DC01

| | |
|---|---|
| **Cause** | Windows Defender Firewall blocks inbound ICMP echo requests by default. |
| **Fix** | None needed; this is the secure default. DC01 had working connectivity through the firewall. Separate connectivity from policy before escalating: a failed ping does not mean a host is down. |